

EE4C5/5C5 Digital Signal Processing

Lecture 4 – Sampling and reconstruction

CA in this module

Prerequisite skills

- MATLAB familiarity for signals and systems
- Proficiency quiz on Blackboard
- Do Lab 0 in your own time if you need it:
 - Clinic on Tues 22nd (see timetable)
 - Optional to attend.

Labs to build skills– Lab A, B, C

- Available to entire class 2 weeks before first associated clinic.
- Spend ~3-4 hours on the assignment BEFORE the scheduled clinic.
 - Gather required plots and observations as you work.
- Come to Clinic to resolve issues preventing you finishing, or to ask questions.
- Attendance at clinic is optional but will be recorded.
- Submission for each Lab NOT a full engineering report.
 - Pdf with figures and observations from each task.
 - Number them by the tasks in handout.
 - All code in separate .m files.
 - Submitted according to the deadline in Blackboard.
 - No marks – formative activity
- Any late submissions must be due to exceptional circumstances or else late policy applied.
 - <https://www.tcd.ie/media/tcd/academic-affairs/pdfs/Exceptional-Circumstances-26-02-16.pdf>

Schedule (subject to change)

Teaching Week	Labs associated tasks for you	Scheduled DML Slots
Week 1	Matlab Quiz & Lab 0 if needed	-
Week 2	Lab A – do in own time	Lab 0 Clinic, optional attendance
Week 3	Lab A – do in own time	-
Week 4		Lab A Clinic in DML, optional attendance
Week 5	Lab B – do in own time	-
Week 6	Lab B – do in own time	-
Week 7 (Reading Week)		-
Week 8	Lab C – do in own time	Lab B Clinic in DML, Optional attendance
Week 9	Lab C – do in own time	-
Week 10	Have completed ALL labs	Lab C Clinic in DML, Optional attendance
Week 11		CA in-person individual assignment (Test)
Week 12		CA in-person individual assignment (Test)

} One of these slots

Late submission policy

- Formative activity.
- Feedback is from your clinic.
- Submission is to show you did the work.
 - Clinic Tuesday, due Friday
- No extensions as work aligned with module delivery.
- Submission portal will remain open until following Tuesday
 - i.e. 1 week after clinic.
 - Late once is OK
 - Late more than once in Semester => lose 3% of the final 15%

Every time you submit on BB

- Will be agreeing to the following:
- I have read and understood the plagiarism provisions in the General Regulations of the University Calendar for the current year, found at <http://www.tcd.ie/calendar>.
- I have also read and understood the guide, and completed the 'Ready Steady Write' Tutorial on avoiding plagiarism, located at <https://libguides.tcd.ie/academic-integrity/ready-steady-write>.
- By submitting this assignment, I am declaring this as my own work. I have not used any AI tools or other third party to write the code for this assignment.

Why would I do labs with no marks?

- The time you spend on this material is meant to additionally reinforce concepts & theory in the module
 - Not just about the CA mark
- Skills built only through the lab activity
 - Clinics to check you are doing it right
- Final session in Week 11/12 will assess skills attained
- Task based on similar ones in Lab A B and C



Lecture 4

This lecture

- Based on (part of) Chapter 4 of O&S
- All images from O&S book unless otherwise stated

Importance of sampling

- Discrete-time signals often represent sampled continuous-time signal
- Ambiguity introduced in the signal
- Potential for aliasing
- How to not lose information?
 - Or at least not the important part
 - Trade-offs

Periodic Sampling

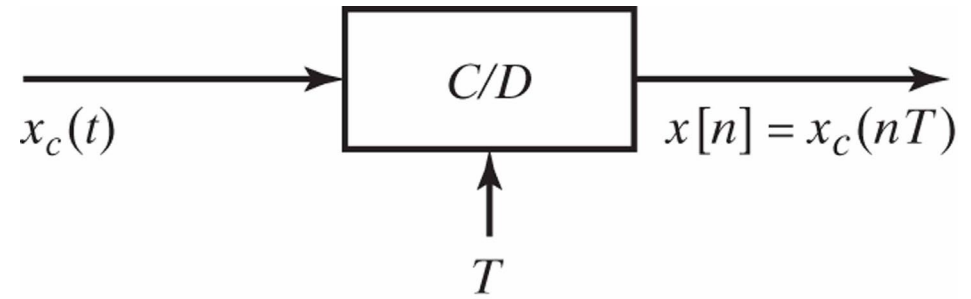
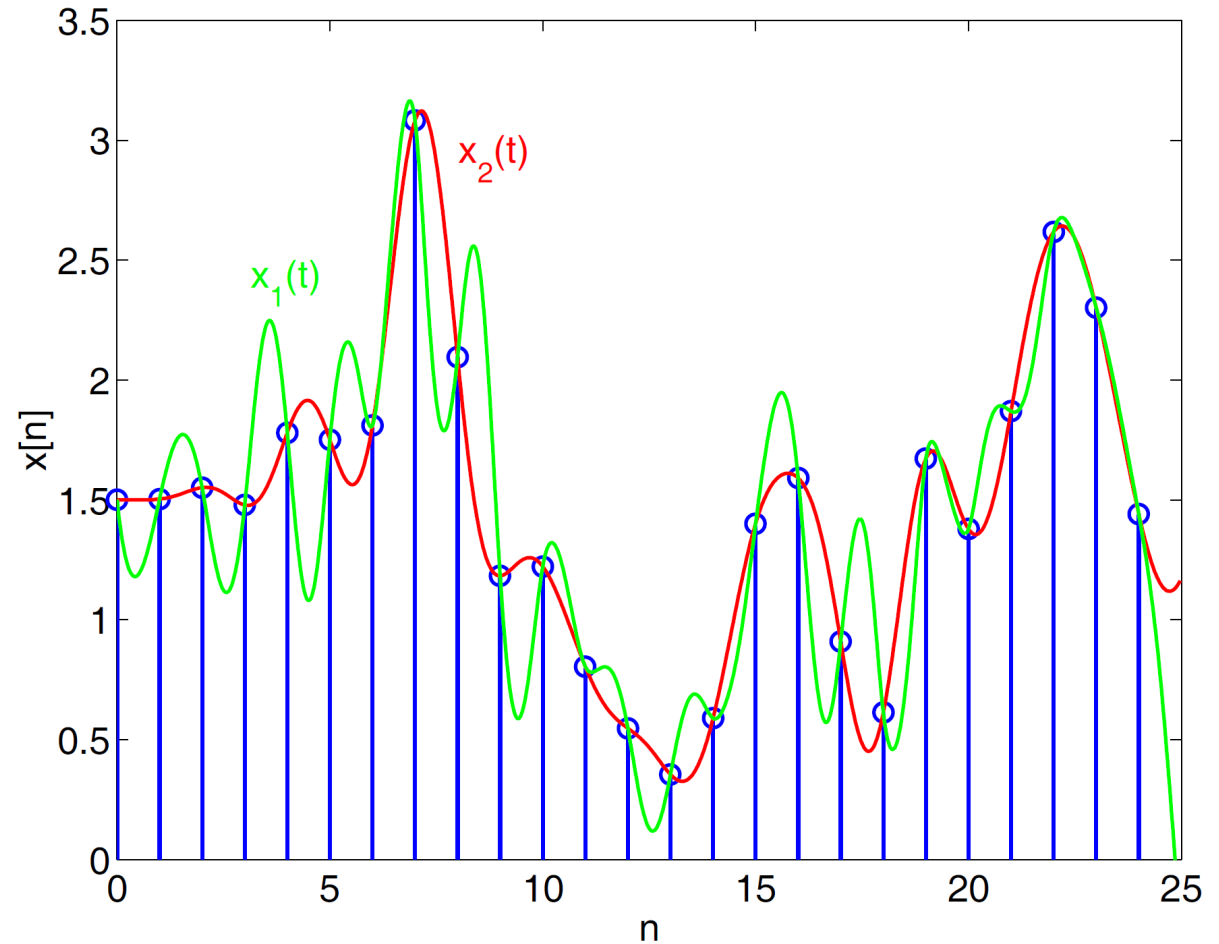


Figure 4.1 Block diagram representation of an ideal continuous-to-discrete-time (C/D) converter.

- Discrete-time representation of a continuous-time signal obtained via periodic sampling
- $x[n] = x_c(nT) \quad -\infty < n < \infty$
- T is sampling period, sampling frequency $f_s = \frac{1}{T}$
- Can also express sampling frequency as $\Omega = \frac{2\pi}{T}$ in radians per second

Many to one...

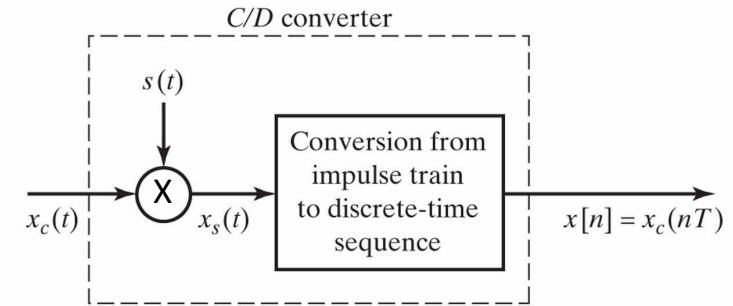
- Ambiguity in discrete-time representation
- Sampled signal $x[n]$ could come from $x_1(t)$ or $x_2(t)$
- Not invertible



Source: Ian Bruce lectures, McMaster University

Impulse train sampling

- Consider as two-stage process
 - Impulse train modulator
 - Conversion to a sequence
- Periodic impulse train $s(t)$
 - $s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$
- With $\delta(t)$ the unit-impulse function
- $x_s(t) = x_c(t) \sum_{n=-\infty}^{\infty} \delta(t - nT)$
- $= \sum_{n=-\infty}^{\infty} x_c(t) \delta(t - nT)$
- $= \sum_{n=-\infty}^{\infty} x_c(nT) \delta(t - nT)$
- Note that $x_s(t)$ is continuous!



(a)

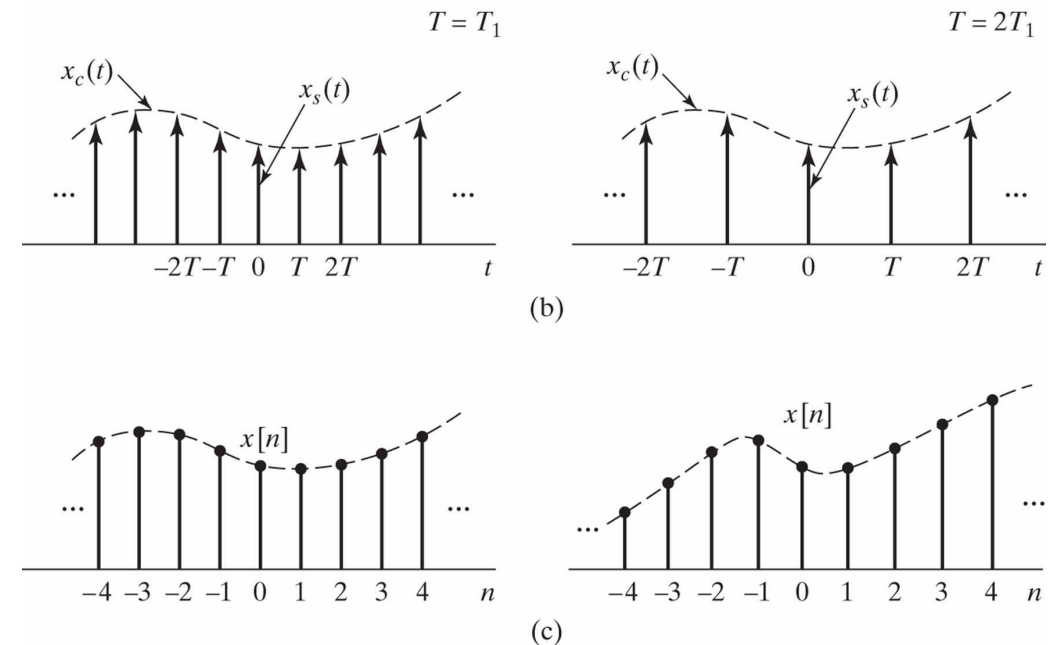


Figure 4.2 Sampling with a periodic impulse train, followed by conversion to a discrete-time sequence. (a) Overall system. (b) $x_s(t)$ for two sampling rates. (c) The output sequence for the two different sampling rates.

Frequency Domain?

- Fourier transform of the periodic impulse train $s(t)$ is a periodic pulse train!
 - $S(j\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s)$
- with $\Omega_s = \frac{2\pi}{T}$ the sampling frequency in radians/s
- Since
 - $X_s(j\Omega) = \frac{1}{2\pi} X_c(j\Omega) * S(j\Omega)$
- Then with the continuous convolution above becomes
 - $X_s(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$

What does this mean??

- $X_s(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$
- Original $X_c(j\Omega)$ bandlimited such that $X_c(j\Omega) = 0$ for $|\Omega| > \Omega_N$
- $X_s(j\Omega)$ is periodically repeated copies shifted by the sampling frequency
- Need $\Omega_s - \Omega_N \geq \Omega_N$

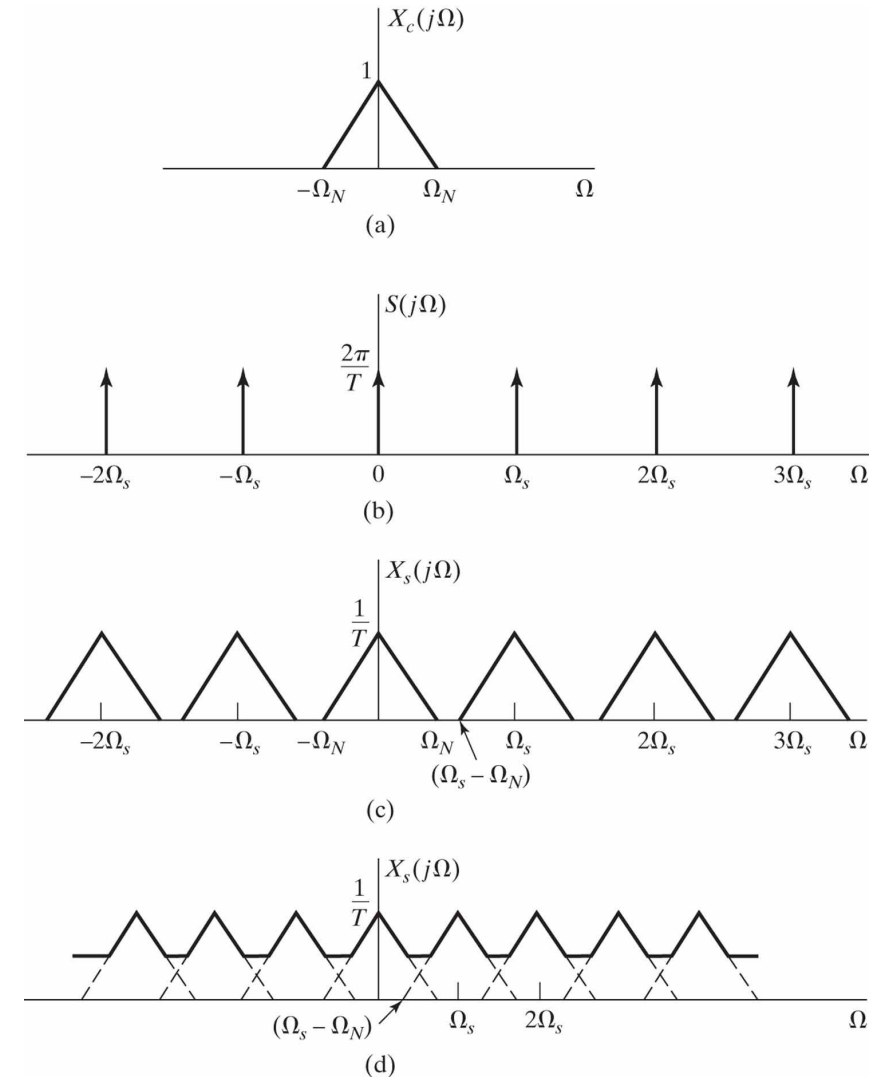


Figure 4.3 Frequency-domain representation of sampling in the time domain. (a) Spectrum of the original signal. (b) Fourier transform of the sampling function. (c) Fourier transform of the sampled signal with $\Omega_s \geq 2\Omega_N$. (d) Fourier transform of the sampled signal with $\Omega_s < 2\Omega_N$.

Nyquist Theorem

- Nyquist–Shannon sampling theorem
- If $x_c(t)$ is a bandlimited signal with $X_c(j\Omega) = 0$ for $|\Omega| > \Omega_N$
- Then $x_c(t)$ is uniquely determined by its samples $x[n] = x_c(nT), n = 0, \pm 1, \pm 2, \dots$, if
- $\Omega_s = \frac{2\pi}{T} \geq 2\Omega_N$
- So to avoid aliasing, the sampling rate must be at least twice the bandwidth of the signal you are sampling

Terminology

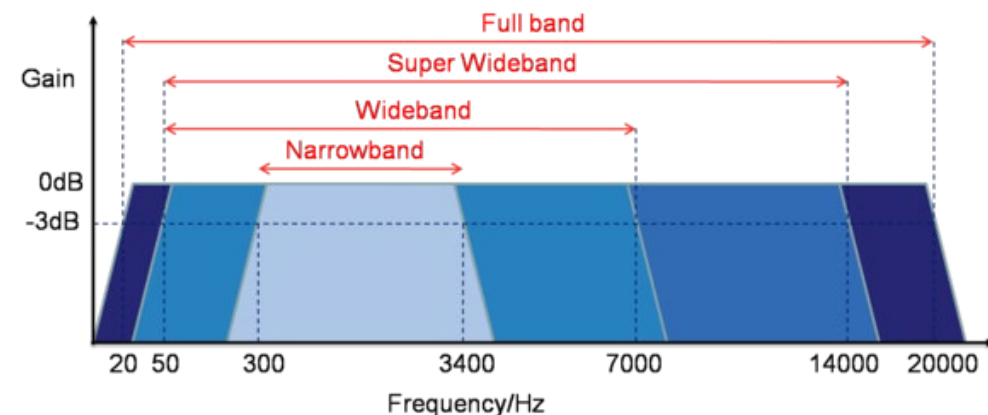
- Ω_N the Nyquist Frequency
- $2\Omega_N$ the Nyquist Rate

Effect of Sampling in the Frequency Domain

- Oversampling
 - The sampling frequency is higher than the Nyquist rate
- Undersampling
 - The sampling frequency is lower than the Nyquist rate
- Critical sampling
 - The sampling frequency is equal to the Nyquist rate
- Note: A pure sinusoid may not be recoverable from its critically sampled version

Effect of Sampling in the Frequency Domain

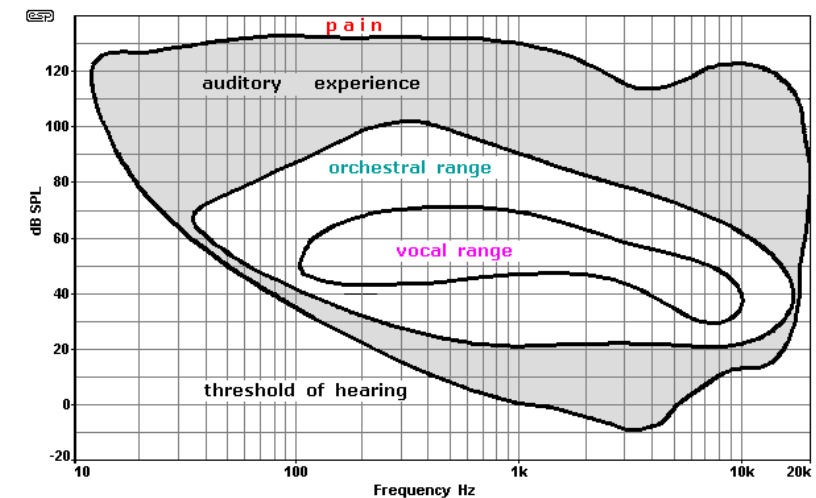
- In digital telephony (“landline”), a 3.4 kHz signal bandwidth is acceptable for telephone conversation
- Here, a sampling rate of 8 kHz, which is greater than twice the signal bandwidth, is used



https://blog.tmcnet.com/voice-of-ip/2010/03/hd_voice_-_how_much_bandwidth_do_you_need_1.html

Effect of Sampling in the Frequency Domain

- In high-quality analog music signal processing, a bandwidth of 20 kHz has been determined to preserve the fidelity
- Hence, in compact disc (CD) music systems, a sampling rate of 44.1 kHz, which is slightly higher than twice the signal bandwidth, is used



https://people.ece.cornell.edu/land/courses/ece5030/FinalProjects/s2014/kkp37_rjs483/kkp37_rjs483/AudioGram.html

Reconstruction

- Can recover original signal with an ideal lowpass filter (note scaling)
- $X_r(j\Omega) = H_r(j\Omega) X_s(j\Omega)$
- If $\Omega_N \leq \Omega_c \leq \Omega_s - \Omega_N$
- Then
- $X_r(j\Omega) = X_c(j\Omega)$
- Works in theory!

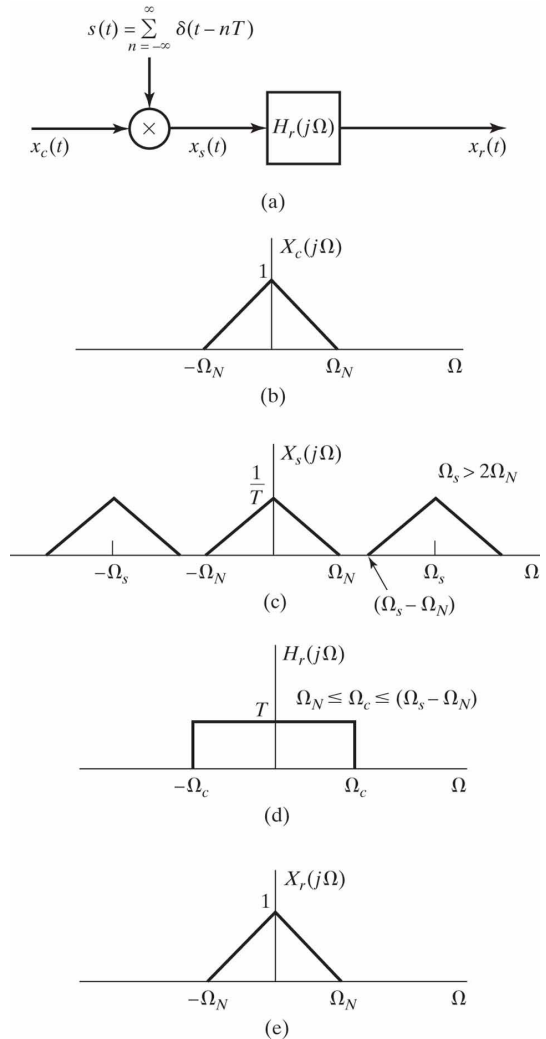


Figure 4.4 Exact recovery of a continuous-time signal from its samples using an ideal lowpass filter.

Reconstruction of signal from samples

- Form an impulse train $x_s(t)$ from a sequence $x[n]$
 - $x_s(t) = \sum_{n=-\infty}^{\infty} x[n] \delta(t - nT)$
- Pass $x_s(t)$ through ideal lowpass filter with frequency response $H_r(j\Omega)$ and corresponding impulse response $h_r(t)$ (inverse FT of $H_r(j\Omega)$)
 - $x_r(t) = \sum_{n=-\infty}^{\infty} x[n] h_r(t - nT)$
- With cut-off frequency $\frac{\pi}{T}$, impulse response given by
 - $h_r(t) = \frac{\sin \pi t/T}{\pi t/T}$

Interpolation

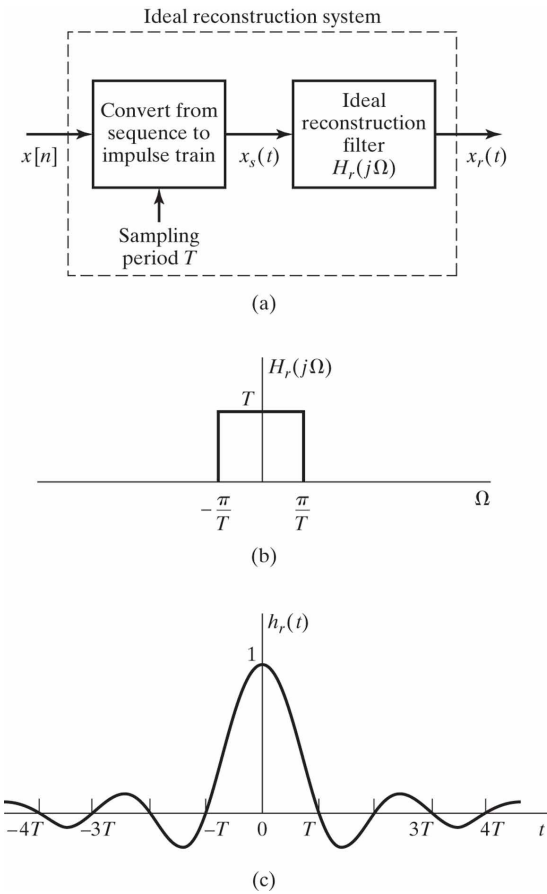


Figure 4.7 (a) Block diagram of an ideal bandlimited signal reconstruction system. (b) Frequency response of an ideal reconstruction filter. (c) Impulse response of an ideal reconstruction filter.

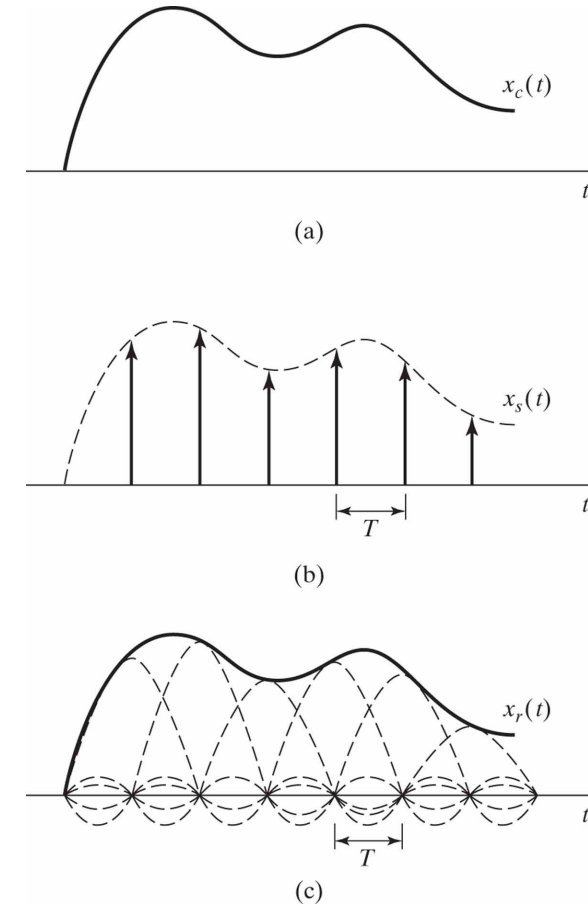


Figure 4.8 Ideal bandlimited interpolation.

Nyquist Theorem in the real world?

- Obey Nyquist => get perfect reconstruction?
- Mathematically you can prove you can recover the signal from the samples exactly
- In reality, this is the situation:
 - Real signals are not strictly bandlimited
 - An ideal “brick-wall” LPF doesn’t exist
 - Sampling introduces noise (ADC)
 - Finite precision effects add to difficulty in realising exact filters
- Best we can hope for?
 - If the signal well band-limited, sampling rate well above Nyquist, well-designed filters, the reconstructed signal will be close enough to the original that we can use it in our application

Required Reading & other material

- Oppenheim & Schafer, Chapter 4, up to end Section 3